



JABchem



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Past Papers Advanced Higher Chemistry

2024 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/160)	%	
A	104+	65%	26.8%
B	88+	55%	22.2%
C	72+	45%	21.9%
D	56+	35%	16.5%
No award	<56	<35%	12.6%

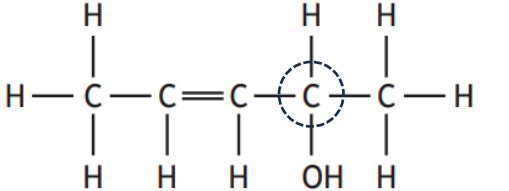
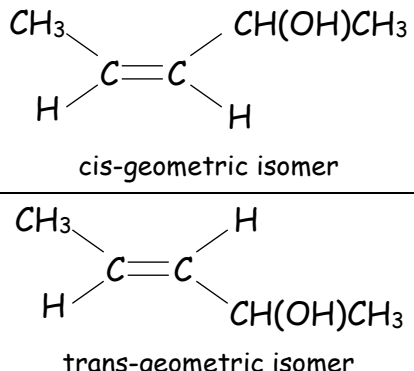
Section:	Multiple Choice	Extended Answer	Project
Average Mark:	15.0 /25	47.0 /95	24.3 /40

2024 Adv Higher Chemistry Marking Scheme

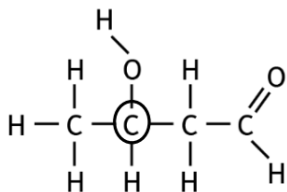
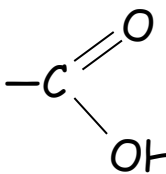
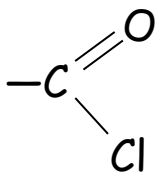
MC Qu	Answer	% Correct	Reasoning																											
1	D	65	☒A The outermost occupied orbital is helium is an s orbital but not the other noble gases ☒B The outermost occupied orbital is an p orbital in all noble gases except helium ☒C Transition metals fill up d-orbitals not noble gases ☒D Helium fills up an s-orbital and the other noble gases fill up a p-orbital																											
2	C	71	Copper atoms have electronic configuration of $4s^1$ in the outer shells one electron is borrowed from 4s to completely fill the 3d orbitals of copper. Ionisation removes 4s electrons before the removal of 3d electrons. <table><tr><td>Species</td><td>Cu</td><td>Cu⁺</td><td>Cu²⁺</td></tr><tr><td>Electronic Configuration</td><td>$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$</td><td>$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$</td><td>$1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$</td></tr><tr><td>Answer</td><td>D</td><td>B</td><td>A</td></tr></table>	Species	Cu	Cu ⁺	Cu ²⁺	Electronic Configuration	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$	Answer	D	B	A															
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Answer	D	B	A																											
3	B	63	<table><tr><td>Oxidation Number in VO²⁺</td><td>Oxidation Number of Vanadium on [V(OH₂)₆]²⁺</td></tr><tr><td>If O=-2 and ion has 2+ charge ∴ V= +4</td><td>OH₂ is neutral so V²⁺ ion ∴ V = +4</td></tr><tr><td colspan="2">∴ Reduction of oxidation of V from 4 to 2 = reduction</td></tr></table>	Oxidation Number in VO ²⁺	Oxidation Number of Vanadium on [V(OH ₂) ₆] ²⁺	If O=-2 and ion has 2+ charge ∴ V= +4	OH ₂ is neutral so V ²⁺ ion ∴ V = +4	∴ Reduction of oxidation of V from 4 to 2 = reduction																						
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4	B	30	Greatest Energy gap between split d-orbitals would be indicated by the colour of the light absorbed have the greatest energy and lowest wavelength. ☒A Purple colour emitted ∴ green wavelengths absorbed (wavelength = 500-560nm) ☒B Green colour emitted ∴ purple wavelengths absorbed (wavelength = 380-400nm) ☒C Blue-Green colour emitted ∴ red wavelengths absorbed (wavelength = 605-750nm) ☒D Red colour emitted ∴ blue-green wavelengths absorbed (wavelength = 490-500nm)																											
5	D	56	tetraamminedichloridocobalt(III) = [CoCl₂(NH₃)₄]⁺ no. of ligands NH₃ ligand no. of ligands Cl⁻ ligand metal name charge on metal ion <table><tr><td colspan="2">Neutral ligands include:</td><td colspan="2">Negative Ligands include:</td><td>Central Ion:</td><td>Charge:</td></tr><tr><td>Ligand</td><td>Name</td><td>Ligand</td><td>Name</td><td>Positive Complex: metals keep their name</td><td rowspan="4">Charge of central ion is converted into roman numerals and put in brackets</td></tr><tr><td>H₂O</td><td>aqua</td><td>Chloride Cl⁻</td><td>chlorido</td><td>Negative Complex: Metals end in ATE</td></tr><tr><td>NH₃</td><td>ammine</td><td>Cyanide CN⁻</td><td>cyanido</td><td>e.g. Cuprate, Ferrate, Cobaltate</td></tr><tr><td>CO</td><td>carbonyl</td><td>Nitrite NO₂⁻</td><td>nitrito</td><td></td></tr></table> NB: Ligands are listed in alphabetical order ∴ ammine listed before chloride Co ion has 3+ charge as Complex Ion has 1+ charge overall and 2- charges from 2xCl⁻ ions	Neutral ligands include:		Negative Ligands include:		Central Ion:	Charge:	Ligand	Name	Ligand	Name	Positive Complex: metals keep their name	Charge of central ion is converted into roman numerals and put in brackets	H ₂ O	aqua	Chloride Cl ⁻	chlorido	Negative Complex: Metals end in ATE	NH ₃	ammine	Cyanide CN ⁻	cyanido	e.g. Cuprate, Ferrate, Cobaltate	CO	carbonyl	Nitrite NO ₂ ⁻	nitrito	
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6	C	25	Electron Pairs = $\frac{\text{No. of electrons in outer shell of central atom} + \text{No. of bonds} - \text{Charge}}{2}$ Electron Pairs = $\frac{8 + 2 - 0}{2}$ Electron Pairs = $\frac{10}{2}$ = 5 pairs = Trigonal bipyramidal electron pair arrangement 5 electron pairs form trigonal bipyramidal shape. Bonding pairs of electrons in Xe-F bonds take the top and bottom positions to allow the lone pairs of electrons to be 120° apart in the trigonal position in middle of shape. XeF₂ is linear in shape as a result. lone pair - lone pair repulsion > lone pair - bonding pair repulsion > bonding pair - bonding pair repulsion																											

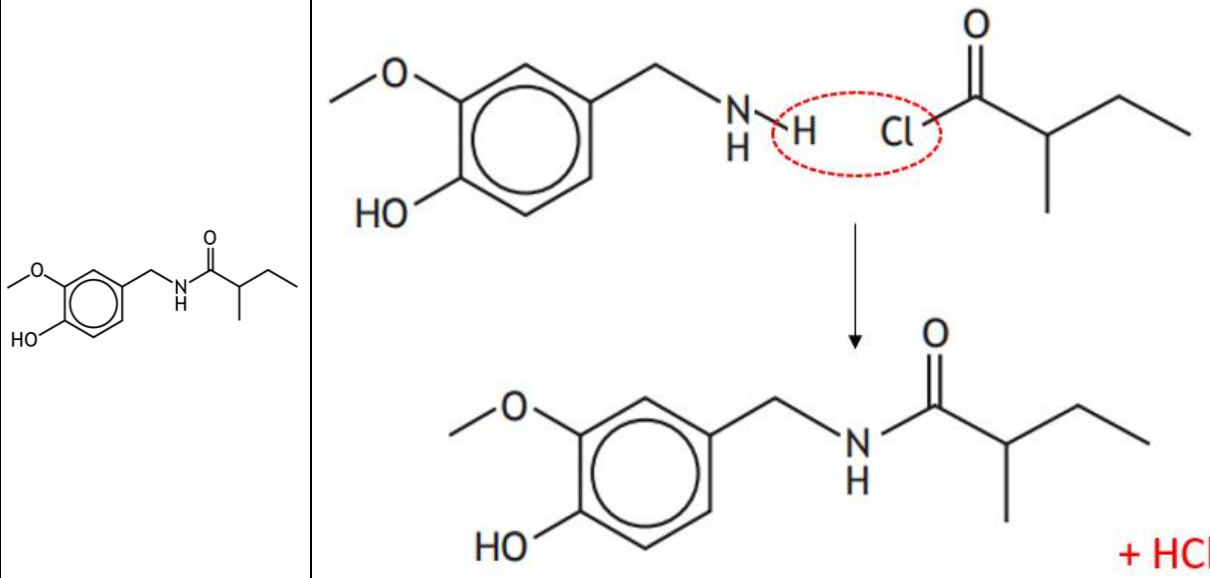
7	B	49	$\underset{\text{propanoic acid}}{\text{C}_2\text{H}_5\text{COOH}} + \underset{\text{water}}{\text{H}_2\text{O}} \rightleftharpoons \underset{\text{propanoate ions}}{\text{C}_2\text{H}_5\text{COO}^-} + \underset{\text{hydronium ion}}{\text{H}_3\text{O}^+}$ adding propanoate ions will shift equilibrium to left: ☒ A equilibrium shifting to left <i>maintains</i> the concentration of propanoate ions at equilibrium ☒ B equilibrium shifting to left <i>decreases</i> H_3O^+ ion concentration ☒ C reduction in H_3O^+ ions concentration which would <i>increase</i> OH^- ion concentration ☒ D equilibrium shifting to left <i>increases</i> the concentration of propanoic acid molecules
8	B	40	$K_w = [\text{H}_3\text{O}^+] \times [\text{OH}^-] = 5.48 \times 10^{-14}$ $[\text{H}_3\text{O}^+]^2 = 5.48 \times 10^{-14}$ $[\text{H}_3\text{O}^+] = 2.34 \times 10^{-7}$ $\log_{10}[\text{H}_3\text{O}^+] = -6.63$ $\text{pH} = -\log_{10}[\text{H}_3\text{O}^+] = 6.63$
9	A	55	☒ A the same volume and concentration of hydrochloric and methanoic acids will neutralise the same number of moles of calcium carbonate at the end of the reaction. Both acids release one H^+ ion to become H_3O^+ ion in water. ☒ B methanoic acid will take longer to complete the reaction as it is a weak acid ☒ C calcium methanoate salt formed is alkaline and will alter the pH at end of reaction ☒ D methanoic acid is a weak acid and will have a decreased initial rate of reaction
10	D	83	$\text{pH} = \text{pK}_a - \log_{10} \frac{[\text{Acid}]}{[\text{Salt}]}$ $\text{pH} = 4.76 - \log_{10} \frac{0.1}{0.1}$ $\text{pH} = 4.76 - \log_{10} (1)$ $\text{pH} = 4.76 - 0$ $\text{pH} = 4.76$
11	A	79	☒ A Reactions with a negative ΔG° value are feasible reactions at that temperature ☒ B The ΔH enthalpy change of the reaction cannot be calculated from the information given ☒ C The moles of reactants in the rate determining step is required to work out the overall order of a reaction. The reaction given is the overall reaction not the r.d.s. ☒ D The size of the activation energy cannot be concluded from the information given
12	C	51	☒ A Order increases on formation of <i>solid</i> snow from <i>liquid</i> water giving a negative value of ΔS ☒ B Order increases on formation of <i>solid</i> snow from <i>liquid</i> water giving a negative value of ΔS ☒ C Decrease in disorder of snowflake = $-\Delta S$ and increase in disorder of surroundings = $+\Delta S$ ☒ D Heat energy released on freezing would increase the disorder of surrounding (positive ΔS)
13	A	47	The gradient of the line is constant $\therefore$ rate of reaction is constant Rate of reaction is constant when concentration of reactant A is changing $\therefore$ zero order kinetics
14	A	59	☒ A The only structure of CH_2O is an aldehyde (methanal) ☒ B $\text{C}_2\text{H}_4\text{O}$ could be an aldehyde (ethanal) or an alkenol (ethenol) ☒ C $\text{C}_2\text{H}_6\text{O}$ could be an alcohol (ethanol) or an ether (methoxymethane) ☒ D $\text{C}_3\text{H}_6\text{O}$ could be an aldehyde (propanal) or a ketone (propanone)
15	B	40	☒ A Acidified permanganate would oxidise propanal ($\text{C}_2\text{H}_5\text{CHO}$) to propanoic acid ($\text{C}_2\text{H}_5\text{COOH}$) ☒ B Fehling's solution reacts with aldehydes not primary/secondary alcohols like $\text{C}_3\text{H}_7\text{OH}$ ☒ C The hydrolysis of the ester $\text{C}_2\text{H}_5\text{COOC}_2\text{H}_5$ would form propanoic acid $\text{C}_2\text{H}_5\text{COOH}$ ☒ D Acid hydrolysis of $\text{C}_2\text{H}_5\text{CN}$ would produce propanoic acid ($\text{C}_2\text{H}_5\text{COOH}$)

16	D	56	☒ A Alcohols can be used as a polar solvents and ethers can be used as non-polar solvents ☒ B Alcohols and ethers in water form hydrogen bonds with water (NB pure ethers do not) ☒ C Alcohols can be made from nucleophilic substitution of haloalkanes with hydroxide ions Ethers can be made from nucleophilic substitution of haloalkanes with alkoxide ions e.g. $C_2H_5O^-$ ☒ D peak at 3600cm^{-1} and 3200cm^{-1} is hydrogen bonded O - H stretch in alcohols but not ethers									
17	C	45	☒ A Amine $-NH_2$ group reacts with acids but aldehyde $-CHO$ group does not react with alkali ☒ B Carboxyl $-COOH$ group reacts with alkalis but hydroxyl $-OH$ group does not react with acids ☒ C Amine $-NH_2$ group reacts with acids and carboxyl $-COOH$ group reacts with alkali ☒ D Amine $-NH_2$ group reacts with acids but hydroxyl $-OH$ group does not react with alkali									
18	D	45	☒ A Only mechanism for reaction 1 contains the formation of a cyclic ion intermediate ☒ B Only mechanism for reaction 2 contains the formation of a carbocation ☒ C Only mechanism for reaction 1 contains the breaking of a pi bond ☒ D Reaction 1 has heterolytic fission of bond in Br_2 where both electrons go to same end Reaction 2 has heterolytic fission of the C - Br bond where both electrons in bond leave to form a Br^- ion.									
19	B	65	H CH₃ C=C H₃C CH₃ 2-methylbut-2-ene $CH_3CH=C(CH_3)_2$ $\longrightarrow$ H C=O H₃C ethanal + O=C CH₃ CH₃ propanone									
20	A	67	☒ A HNO_3/H_2SO_4 mixture produces the electrophile species Nitronium ion NO_2^+ and $-NO_2$ group is substituted onto benzene ring. ☒ B conc H_2SO_4 alone will substitute SO_3H groups onto benzene ring. ☒ C The substitution of groups onto a benzene ring is always electrophilic not nucleophilic ☒ D The substitution of groups onto a benzene ring is always electrophilic not nucleophilic									
21	C	83	<table><tr><td>Molecule</td><td>X</td><td>Y</td></tr><tr><td>Structure</td><td></td><td></td></tr><tr><td>Naming of geometric isomer</td><td><u>cis-isomer</u> Both bromines are above the plane of fixed carbon ring</td><td><u>trans-isomer</u> bromines are arranged across the plane of the fixed carbon ring</td></tr></table> ☒ A cis-isomer has Br groups on same side of ring and trans-isomer has Br groups on opposite sides ☒ B molecules shown are numbered 1,2-dibromocyclopropane to give a lower numbering system ☒ C Numbering of ring correct and identification of cis- & trans- isomers correct. ☒ D molecules shown are numbered 1,2-dibromocyclopropane to give a lower numbering system	Molecule	X	Y	Structure			Naming of geometric isomer	<u>cis-isomer</u> Both bromines are above the plane of fixed carbon ring	<u>trans-isomer</u> bromines are arranged across the plane of the fixed carbon ring
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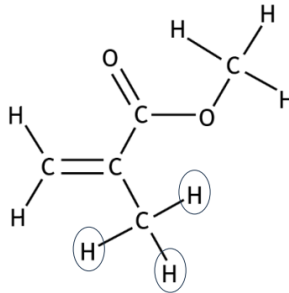
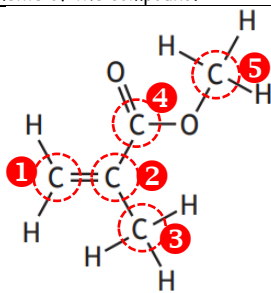
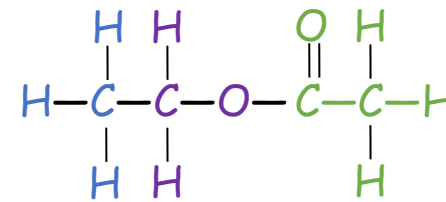
22	D	26	2 optical isomers from:  Circled carbon is a chiral carbon as it has four different groups attached to it. This leads to two stereoisomers which rotate plane polarised light different directions	2 geometric isomers from:  cis-geometric isomer trans-geometric isomer
23	A	46	☒ A Protons flip in ^1H NMR and energy is emitted when peaks are produced ☒ B It is protons that flip when radiation is applied in ^1H NMR (aka proton NMR) ☒ C The peaks are produced when energy is emitted ☒ D It is protons that flip when radiation is applied in ^1H NMR (aka proton NMR)	
24	C	63	gfm $\text{H}_2\text{S}_2\text{O}_3 = (2 \times 1) + (2 \times 32.1) + (3 \times 16) = 2 + 64.2 + 48 = 114.2\text{g}$ $\% \text{ mass S} = \frac{\text{mass of S}}{\text{gfm}} \times 100 = \frac{2 \times 32.1}{114.2} \times 100 = 56.2\%$ 56.2% of total mass = 5.182g 100% of total mass = $5.182\text{g} \times \frac{100}{56.2}$ = 9.218g $\% \text{ mass O} = \frac{\text{mass of O}}{\text{gfm}} \times 100 = \frac{3 \times 16}{114.2} \times 100 = 42.0\%$ Mass of O in sample = 42.0% of 9.218g = $\frac{42.0}{100} \times 9.218\text{g} = 3.874\text{g}$	
25	A	84	☒ A All antibiotics contain all groups to the right of (and including) the $\text{C}=\text{O}$ carbonyl group ☒ B Methicillin is does not contain anything to the left of the $\text{C}=\text{O}$ carbonyl group ☒ C Methicillin is does not contain anything to the left of the $\text{C}=\text{O}$ carbonyl group ☒ D Methicillin is does not contain anything to the left of the $\text{C}=\text{O}$ carbonyl group	

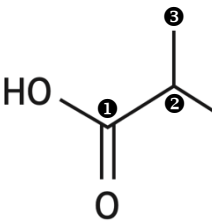
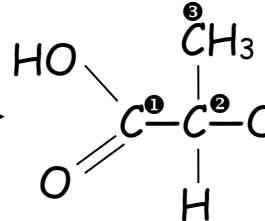
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Long Qu	Answer	Reasoning			
1a(i)	2 nd order or 2	rate = k x [CH ₃ CHO] x [OH ⁻] ∴ rate = k x [CH ₃ CHO] ¹ x [OH ⁻] ¹ Order of CH ₃ CHO = 1 Order of OH ⁻ = 1 ∴ Overall Order = 1 + 1 = 2			
1a(ii)	1.15 l mol ⁻¹ s ⁻¹ (1mark) (1mark)	$k = \frac{\text{rate}}{[\text{CH}_3\text{CHO}][\text{OH}^-]} = \frac{1.72 \times 10^{-3} \text{ mol l}^{-1} \text{ s}^{-1}}{0.100 \text{ mol l}^{-1} \times 0.0150 \text{ mol l}^{-1}} = 1.15 \text{ l mol}^{-1} \text{ s}^{-1}$			
1a(iii)	Step 1 with one explanation from:	Both reactants/particles/species are involved	CH ₃ CHO and OH ⁻ are involved	Step 1 is first order for both reactants	Step 1 matches the rate equation
1b(i)		A chiral centre or chiral carbon is a carbon with four different groups attached to it which leads to optical isomerism.			
1b(ii)	one answer from:	Contains equal amounts of both isomers/ enantiomers		It is a racemic mixture	
2a(i)	one answer from:	Unreactive (with substances in the chilli peppers)	Volatile or Low boiling point	Able to dissolve the capsaicinoid	Similar polarity to capsaicinoid
2a(ii)	one answer from:	Vacuum		Under reduced pressure	
2b	140070	2.00g dried chilli peppers contains 17.4mg capsaicinoids 0.002 kg dried chilli peppers contains 17.4mg capsaicinoids 1 kg dried chilli peppers contains 17.4mg capsaicinoids x ¹ / _{0.002} = 8700mg capsaicinoids ∴ Concentration = 8700 mg in 1kg = 8700ppm Scoville heat value = 16.1 x Concentration in ppm = 16.1 x 8700 = 140070			
2c(i)	Agonist	agonist	An agonist mimics the natural compound and binds to the receptor molecules to produce a response similar to the natural active compound.		
		antagonist	An antagonist prevents the natural compound from binding to the receptor, and so blocks the natural response from occurring.		
2c(ii)	0.0825	gfm Capsaicin = 305g mass = no. of mol x gfm = 1.15x10 ⁻⁴ x 305 = 0.0351g $\% \text{ mass} = \frac{\text{mass of capsaicin}}{\text{mass of cream}} \times 100 = \frac{0.0351\text{g}}{42.5} \times 100 = 0.0825\%$			
2d(i)	Acid chloride	Functional Group found in Carboxylic Acids 		Functional Groups found in Acid Chloride 	
2d(ii)	Condensation or nucleophilic substitution	A condensation reaction happens when two smaller molecules join together and a small molecule is removed in the joining up of the molecules. In this reaction HCl is removed in the condensation process.			

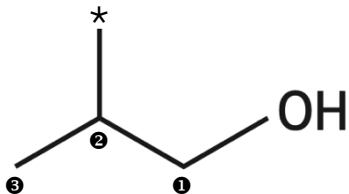
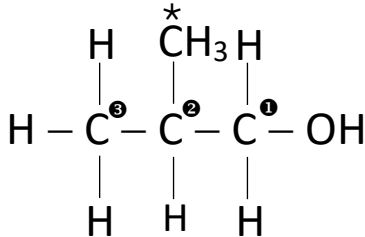
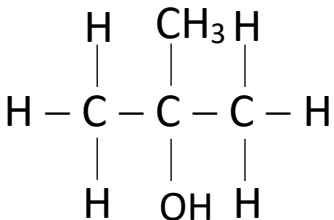
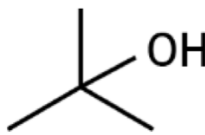
2d(iii)				
3a(i)	One answer from: Electrons in the same $\left\{ \begin{array}{c} \text{box} \\ \text{orbital} \end{array} \right\}$ have $\left\{ \begin{array}{c} \text{opposite} \\ \text{different} \end{array} \right\}$ spin Electrons are paired with opposite spins in each $\left\{ \begin{array}{c} \text{box} \\ \text{orbital} \end{array} \right\}$			
3a(ii)	5 0 0 $+\frac{1}{2}$ or 5 0 0 $-\frac{1}{2}$ Principal Quantum number (n) = 5 as electrons is in 5th shell Angular Momentum Quantum Number (l) = 0 as it is an s-subshell for 5s $s = 0$, $p = 1$, $d = 2$ and $f = 3$ Magnetic Quantum Number (m_l) = 0 as s subshells only have value $m_l = 0$ p can be -1,0,+1, d can be -2,-1,0,+1,+2 and f can be -3,-2,-1,0,+1,+2,+3 Spin Quantum Number (m_s) has only two values: $+\frac{1}{2}$ or $-\frac{1}{2}$			
3b(i)	electron falls/moves down or jumps/moves to lower level Electrons are excited and jump to a level higher than the ground state. An emission spectrum for that element is obtained when the electrons fall back to a lower level, the electrons release specific quantities of energy which is dependent on the starting and final energy levels the electron jumped across. This difference in energy is quite specific and different from atom to atom. The energy that is released has particular wavelengths which correspond to the exact quantities of energy released.			
3b(ii)	477 $E = \frac{L \times h \times c}{\lambda} \therefore \lambda = \frac{L \times h \times c}{E} = \frac{6.02 \times 10^{23} \text{ mol}^{-1} \times 6.63 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ m s}^{-1}}{251 \times 1000 \text{ J mol}^{-1}}$ $= 4.77 \times 10^{-7} \text{ m}$ $= 477 \text{ nm}$			
3c(i)	2 The strontium Sr^{2+} ions form two dative covalent bonds with pairs of electrons in oxygen atoms giving each strontium Sr^{2+} ions a co-ordination number of 2			
3c(ii)	tetradentate The ranelate ion is tetradentate as it is able to donate four pairs of electrons over the whole ligand to metal ions.			
3d(i)	one answer from: <table><tr><td>Weigh an accurate mass that is close to 2.5 g</td><td>The exact mass should be known/measured/weighed/given to 2 dp or more and should be close to 2.5 g/desired mass</td><td>The mass does not need to be exactly 2.5 g/desired mass but it must be accurately measured/known/weighed/given to 2 dp or more</td></tr></table>	Weigh an accurate mass that is close to 2.5 g	The exact mass should be known/measured/weighed/given to 2 dp or more and should be close to 2.5 g/desired mass	The mass does not need to be exactly 2.5 g/desired mass but it must be accurately measured/known/weighed/given to 2 dp or more
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3d(ii)	Answer to include:<table><tr><td>Marks Awarded:</td><td>Points Included:</td></tr><tr><td>2 marks</td><td>all 4 points</td></tr><tr><td>1 mark</td><td>2 or 3 points</td></tr><tr><td>0 marks</td><td>1 or less points</td></tr></table>	Marks Awarded:	Points Included:	2 marks	all 4 points	1 mark	2 or 3 points	0 marks	1 or less points	<table><tr><td>heat</td><td>cool in a desiccator</td><td>weigh (crucible/sample)</td><td>repeat (heating, cooling and weighing) until there is no further change in mass / constant mass</td></tr></table>	heat	cool in a desiccator	weigh (crucible/sample)	repeat (heating, cooling and weighing) until there is no further change in mass / constant mass			
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0 marks	1 or less points																
heat	cool in a desiccator	weigh (crucible/sample)	repeat (heating, cooling and weighing) until there is no further change in mass / constant mass														
3d(iii)	6	Mass of $\text{SrCl}_2 \cdot n\text{H}_2\text{O}$ mass of crucible and $\text{SrCl}_2 \cdot n\text{H}_2\text{O}$ - mass of crucible = $17.58 - 14.87 = 2.71\text{g}$ Mass of SrCl_2 mass of crucible and SrCl_2 - mass of crucible = $16.49 - 14.87 = 1.62\text{g}$ Mass of H_2O mass of $\text{SrCl}_2 \cdot n\text{H}_2\text{O}$ - mass of SrCl_2 = $2.71 - 1.62 = 1.09\text{g}$ <table><tr><td>Substance</td><td>SrCl_2</td><td>H_2O</td></tr><tr><td>mass</td><td>1.62g</td><td>1.09g</td></tr><tr><td>gfm</td><td>$(1 \times 87.6) + (2 \times 35.5) = 158.6\text{g}$</td><td>$(2 \times 1) + (1 \times 16) = 18\text{g}$</td></tr><tr><td>no. of moles</td><td>0.01021mol</td><td>0.06056mol</td></tr><tr><td>Mole Ratio</td><td>1 mol</td><td>$0.06056\text{mol} \times \frac{1}{0.01021} = 5.93 = 6\text{mol}$</td></tr></table>	Substance	SrCl_2	H_2O	mass	1.62g	1.09g	gfm	$(1 \times 87.6) + (2 \times 35.5) = 158.6\text{g}$	$(2 \times 1) + (1 \times 16) = 18\text{g}$	no. of moles	0.01021mol	0.06056mol	Mole Ratio	1 mol	$0.06056\text{mol} \times \frac{1}{0.01021} = 5.93 = 6\text{mol}$
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4a(i)	Circle Showing:	Solid Line = minimum accepted and dotted line = maximum accepted															
4a(ii)	Curve with peak at maximum between 500-560nm	If flowers of crocus plants are purple, purple wavelengths must be transmitted and wavelengths of green light between 500-560nm must be absorbed by the pigments in the crocus plant.															
4b	Answer to include:	<table><tr><td>1 mark</td><td>One line represents end on overlap (to form a sigma bond)</td></tr><tr><td>1 mark</td><td>One line represents side on/parallel overlap (to form a pi bond)</td></tr></table>	1 mark	One line represents end on overlap (to form a sigma bond)	1 mark	One line represents side on/parallel overlap (to form a pi bond)											
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4c	Answer to include:	Marks	1 st mark		2 nd mark												
		One row from:	one from:	different conjugation different numbers of carbon atoms in conjugated system different chromophore		Different energy gap between HOMO and LUMO											
			one from:	smaller conjugated system fewer (carbon) atoms in conjugated system smaller chromophore		one from: larger energy gap between HOMO and LUMO shorter wavelength of light absorbed											
			one from:	larger conjugated system more (carbon) atoms in conjugated system larger chromophore		one from: smaller energy gap between HOMO and LUMO longer wavelength of light absorbed											
5	Open Ended Question:	3 mark answer		2 mark answer		1 mark answer											
		Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.		Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.											
		<table><tr><th colspan="4">Marks Awarded</th></tr><tr><td>3 marks</td><td>2 marks</td><td>1 mark</td><td>0 marks</td></tr><tr><td>5%</td><td>23%</td><td>38%</td><td>34%</td></tr></table>		Marks Awarded				3 marks	2 marks	1 mark	0 marks	5%	23%	38%	34%		
Marks Awarded																	
3 marks	2 marks	1 mark	0 marks														
5%	23%	38%	34%														
6a(i)	Radio	Radio waves and frequencies are used in NMR Spectroscopy to flip the spins of protons in the nucleus															
6a(ii)	Tetramethylsilane (TMS)	Tetramethylsilane (CH ₃) ₄ Si is the standard reference substance used in NMR Spectroscopy which gives a peak at 0 ppm on the chemical shift axis.															
6b(i)	1 or more circled hydrogens as shown in diagram: 	A peak at 1.9ppm in the data booklet gives the following possible groups:															
		Group		Chemical Shift (ppm)	Reasoning												
		CH ₃ C=C-, RCH ₂ C=C-, R ₂ CHC=C-		2.6-1.6	CH ₃ attached to C=C group	✓											
		CH ₃ C≡C-, RCH ₂ C≡C-, R ₂ CHC≡C-		2.8-1.7	No C≡C bonds in structure	x											
		-C≡CH		3.1-1.8	No C≡C bonds in structure	x											
		ROH		6.0-1.0	No -OH groups in structure	x											
		RNH ₂ , RNH		3.0-1.1	No N atoms in structure	x											
6b(ii)	Lack of free rotation or restricted rotation	C=C double bonds lack rotation of the bonds on the carbon not taking part in the double bonds. Because the two hydrogens are fixed and do not rotate. They are considered different hydrogen environments as a hydrogen will either be on the same side as the CH ₃ group or the same side as the COOCH ₃ group which will influence the hydrogen environments of the compound.															
6b(iii)	5																
6c	Correct Structural Formula of ethyl ethanoate (or methyl propanoate)	Chemical Shift Peak	Type	No of Adjacent Carbons	Integral Curve Height	Group Identified											
		~1.2	Triplet	2 (2 hydrogens on adjacent carbon)	3 (3 hydrogens in group)	CH ₃ - R											
		~2.1	Singlet	0 (0 hydrogens on adjacent carbon)	3 (3 hydrogens in group)	O = C - CH ₃											
		~4.8	Quadruplet	3 (3 hydrogens on adjacent carbon)	2 (2 hydrogens in group)	R - CH ₂ - O -											
		¹³ C NMR has four peaks which indicates the ester has four carbons 															

7a(i)	$C_3H_6O_3$	  $C_3H_6O_3$ skeletal formula structural formula molecular formula									
7a(ii)	4.41	$pH = \frac{1}{2}pK_a - \frac{1}{2}\log_{10}c$ $pH = \frac{1}{2}x - \log_{10}(1.38 \times 10^{-4}) - \frac{1}{2}\log_{10}(1.12 \times 10^{-5})$ $pH = \frac{1}{2}x - (-3.86) - \frac{1}{2}x(-4.95)$ $pH = 1.930 - (-2.475)$ $pH = 4.405$									
7b(i)	Complexometric (titration)	Titrations involving E.D.T.A. (ethylenediaminetetraacetic acid) are complexometric. E.D.T.A. is a chelating agent that forms complexes with metal ions in a 1mol:1mol ratio and the no. of moles of metal ions can be calculated from the no. of moles of E.D.T.A. used up.									
7b(ii)	119	$\text{no. of mol} = \text{volume} \times \text{concentration} = 0.0149_{\text{litres}} \times 0.0200_{\text{mol l}^{-1}} = 0.000298\text{mol}$ $10\text{cm}^3 \text{ milk sample} = 0.000298\text{mol Ca}^{2+} \text{ ions}$ $100\text{cm}^3 \text{ milk sample} = 0.00298\text{mol Ca}^{2+} \text{ ions}$ $\text{gfm Ca} = 40.1\text{g}$ $\text{mass} = \text{no. of mol} \times \text{gfm} = 0.00298 \times 40.1 = 0.119\text{g} = 119\text{mg}$									
8a(i)	$K = \frac{[SO_2]^2[O_2]}{[SO_3]^2}$	For reaction: $aA + bB \rightleftharpoons cC + dD$ $K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$									
8a(ii)	0.0386	$2SO_3 \rightleftharpoons 2SO_2 + O_2$ 2mol 0.25mol (reacted) 2mol 0.25mol (formed) 1mol 0.125mol (formed) $\text{no. of mol } SO_3 \text{ unreacted} = 0.700\text{mol} - 0.25\text{mol} = 0.45\text{mol}$ $K = \frac{[SO_2]^2[O_2]}{[SO_3]^2} = \frac{[0.25]^2 \times [0.125]}{[0.45]^2} = 0.0386$									
8a(iii)	No effect or same value of K	By changing the container to 2 litres, this will half the pressure in the container. At equilibrium, changes to concentration and pressure do not alter the value of the equilibrium constant K and the proportions of reactants and products at equilibrium will be the same as a result. Changes to temperature will alter the equilibrium constant.									
8a(iv)	Answer to include:	<table><tr><td>1st Mark (one from:)</td><td>reaction is endothermic</td><td>reverse reaction is exothermic</td></tr><tr><td>2nd Mark (one from:)</td><td>Equilibrium shifts to the left</td><td>Reverse reaction/reactants favoured</td></tr><tr><td></td><td>$[SO_3]$ increases or more reactant</td><td>$[SO_2]$ and/or $[O_2]$ decreases or less product</td></tr></table>	1 st Mark (one from:)	reaction is endothermic	reverse reaction is exothermic	2 nd Mark (one from:)	Equilibrium shifts to the left	Reverse reaction/reactants favoured		$[SO_3]$ increases or more reactant	$[SO_2]$ and/or $[O_2]$ decreases or less product
1 st Mark (one from:)	reaction is endothermic	reverse reaction is exothermic									
2 nd Mark (one from:)	Equilibrium shifts to the left	Reverse reaction/reactants favoured									
	$[SO_3]$ increases or more reactant	$[SO_2]$ and/or $[O_2]$ decreases or less product									
8b	one answer from:	greater stronger } repulsion by non-bonding pair(s) lone pair(s) } of electrons in sulfur dioxide									
9a	Zinc ion/ Zn^{2+} does not contain an incomplete d-subshell	To be considered a transition metal, a metal must have at least one ion with an incomplete d-subshell. Zinc only forms one ion (Zn^{2+} ion) and it has a complete d-subshell. Zn atom: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$ Zn atom: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$									
9b(i)A	Answer to include: All 4 points = 2marks 2 or 3 points = 1 mark	<table><tr><td>dissolving (the solid) in distilled/deionised water (in a beaker)</td><td>transferring the solutions and rinsings into a volumetric/standard flask</td><td>making up to the (graduation) mark with distilled/deionised water</td><td>stopping and inverting</td></tr></table>	dissolving (the solid) in distilled/deionised water (in a beaker)	transferring the solutions and rinsings into a volumetric/standard flask	making up to the (graduation) mark with distilled/deionised water	stopping and inverting					
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9b(i)B	Answer to include: All 4 points = 2marks 2 or 3 points = 1 mark	<table><tr><td>Standard/accurate solutions or solutions of known concentration (are prepared/used)</td><td>A suitable filter/wavelength</td><td>Mention of a blank or solvent only measurement.</td><td>Absorbance/transmittance of each solution is measured.</td></tr></table>	Standard/accurate solutions or solutions of known concentration (are prepared/used)	A suitable filter/wavelength	Mention of a blank or solvent only measurement.	Absorbance/transmittance of each solution is measured.					
Standard/accurate solutions or solutions of known concentration (are prepared/used)	A suitable filter/wavelength	Mention of a blank or solvent only measurement.	Absorbance/transmittance of each solution is measured.								

9b(ii)	1.3	From graph: when absorbance = 0.28 then concentration = $1.4 \times 10^{-4} \text{ mol l}^{-1}$ 1 litre = 1000 cm^3 contains $1.4 \times 10^{-4} \text{ mol}$ 500 cm^3 contains $0.7 \times 10^{-4} \text{ mol}$ gfm Mn = 54.9 $\therefore$ mass = no. of mol $\times$ gfm = $0.7 \times 10^{-4} \times 54.9 = 0.003843 \text{ g}$ % mass = $\frac{\text{mass of manganese}}{\text{mass of steel}} \times 100 = \frac{0.003843}{0.285} \times 100 = 1.3\%$															
10a(i)	-105	$\begin{aligned} \Delta H^\circ &= \Sigma \Delta H_f^\circ(\text{products}) - \Sigma \Delta H_f^\circ(\text{reactants}) \\ &= (1 \times -52.6) - (1 \times 52.4) + (\tfrac{1}{2} \times 0) \\ &= -52.6 - 52.4 \\ &= -105 \text{ kJ mol}^{-1} \end{aligned}$															
10a(ii)	-79.5	$\begin{aligned} \Delta G_f^\circ &= \Sigma \Delta G_f^\circ(\text{products}) - \Sigma \Delta G_f^\circ(\text{reactants}) \\ &= (1 \times -13.1) - (1 \times 68.2) + (\tfrac{1}{2} \times 0) \\ &= -13.1 - 68.2 \\ &= -81.3 \text{ kJ mol}^{-1} \end{aligned}$ $\Delta S^\circ = \frac{\Delta H^\circ - \Delta G^\circ}{T} = \frac{-105 - (-81.3)}{298} = -0.0795 \text{ kJ K}^{-1} \text{ mol}^{-1} = -79.5 \text{ J K}^{-1} \text{ mol}^{-1}$															
10a(iii)	1320	$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ = 0 \therefore T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{-105 \times 1000 \text{ J mol}^{-1}}{-79.5 \text{ J K}^{-1} \text{ mol}^{-1}} = 1320 \text{ K}$															
10b	heterogeneous	<table><tr><td>Type of Catalyst</td><td colspan="3">Definition</td></tr><tr><td>Heterogeneous</td><td colspan="3">Catalyst is in a <i>different</i> state to the reactants</td></tr><tr><td>Homogeneous</td><td colspan="3">Catalyst is in the <i>same</i> state as the reactants</td></tr></table>				Type of Catalyst	Definition			Heterogeneous	Catalyst is in a <i>different</i> state to the reactants			Homogeneous	Catalyst is in the <i>same</i> state as the reactants		
Type of Catalyst	Definition																
Heterogeneous	Catalyst is in a <i>different</i> state to the reactants																
Homogeneous	Catalyst is in the <i>same</i> state as the reactants																
10c	One answer from:	Addition	Hydration	Oxidation	Hydrolysis												
10d																	
10e	$\text{N}(\text{CH}_2\text{CH}_2\text{OH})_3$	Also accepted: $\text{C}_6\text{H}_{15}\text{NO}_3$ or correct structural formula															
11a(i)	0.31	no. of mol = volume $\times$ concentration = $0.015 \text{ litres} \times 0.1 \text{ mol l}^{-1} = 0.0015 \text{ mol}$ Formula of KHP = $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOK} = \text{C}_8\text{H}_5\text{O}_4\text{K}$ gfm KHP = $(8 \times 12) + (5 \times 1) + (4 \times 16) + (1 \times 39.1) = 204.1 \text{ g}$ mass = no. of mol $\times$ gfm = $0.0015 \times 204.1 = 0.306 \text{ g}$															
11a(ii)	One answer from:	<table><tr><td>Indicator should have a</td><td> range change colour pK_{In} </td><td colspan="2">within the vertical part of the curve</td></tr><tr><td>Indicator should have a</td><td> range change colour pK_{In} </td><td colspan="2">between 6 and 10</td></tr></table>				Indicator should have a	range change colour pK_{In}	within the vertical part of the curve		Indicator should have a	range change colour pK_{In}	between 6 and 10					
Indicator should have a	range change colour pK_{In}	within the vertical part of the curve															
Indicator should have a	range change colour pK_{In}	between 6 and 10															
11a(iii)	Each sample could have a different mass	The separate samples may be weighed accurately but they are not necessarily going to be equal in mass. Different masses will react with different volumes of sodium hydroxide.															
11b(i)	one answer from:	<table><tr><td>Some of the mass weighed out will be impurities</td><td>Not all the mass weighed out will be KHP</td><td colspan="2">More is needed to get the correct mass/moles of KHP</td></tr></table>				Some of the mass weighed out will be impurities	Not all the mass weighed out will be KHP	More is needed to get the correct mass/moles of KHP									
Some of the mass weighed out will be impurities	Not all the mass weighed out will be KHP	More is needed to get the correct mass/moles of KHP															
11b(ii)	1250	At 99.5% purity, 1000ppm contains 2.137g KHP 1000 cm^3 standard solution $\longleftrightarrow$ 2.137g KHP 250 cm^3 standard solution $\longleftrightarrow$ $2.137 \text{ g KHP} \times \frac{250}{1000} = 0.534 \text{ g KHP}$ 0.534g KHP in 250 cm^3 standard solution $\longleftrightarrow$ 1000ppm 0.670g KHP in 250 cm^3 standard solution $\longleftrightarrow$ $1000 \text{ ppm} \times \frac{0.670}{0.534} = 1254 \text{ ppm}$															

12a	2-methylpropan-1-ol or methylpropane-1-ol			=		
12b(i)	One from:	H^+/H_2O Aqueous acid/named aqueous acid		$H^+(aq)$ Dilute acid/named dilute acid	H_3O^+	
12b(ii)	Structure formula of methylpropan-2-ol	 or 				
12b(iii)	One answer from:	less stable { carbocation intermediate formed } in { methylpropan-1-ol X } or more stable { carbocation intermediate } formed in { major product isomer of X methylpropan-2-ol }				
12c(i)	One answer from:	Bond breaks and each atom retains one electron from the bond		Bond breaks evenly	Diagram with correct half headed curly arrows showing a bond breaking	
12c(ii)	One answer from:	Mixture of products formed			Side reactions	
12d	Open Ended Question:	3 mark answer		2 mark answer		1 mark answer
		Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.		Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
		Marks Awarded				
		3 marks	2 marks	1 mark	0 marks	
		9%	15%	21%	55%	